3D CAD modelling

National 5 · CAD & technology · Worksheet

GRAPHICS WITH MR KELLY

Name _____ Class _____ Date _____

By the end of this worksheet I can…

- ☐ describe how to model a component with extrude, revolve and loft, giving all dimensions
- ☐ describe shell, subtract, fillet and chamfer edits in the right order

I DO — Worked example

Read how a full-mark answer is built

A cylindrical pen pot is Ø50 and 50 tall. It has a Ø35 hole 40 deep from the top, a 2.5 × 45° chamfer on the top edge and a R2 fillet on the bottom edge.

1 Describe how the pen pot would be modelled using 3D CAD. (5 marks)

Full-mark answer each ✓ is one mark

Sketch a Ø50 circle on the base work plane ✓ and extrude it 50 ✓. On the top face, sketch a Ø35 circle and extrude-subtract it 40 ✓. Chamfer the top outer edge 2.5 × 45° ✓ and fillet the bottom edge R2 ✓.

Why answers lose marks

"Draw a circle and extrude it. Cut out the middle. Add a chamfer and a fillet."

1 mark at most: no sizes, "cut out" is not a CAD term, and the edges for the chamfer and fillet are not identified.

WE DO — Guided practice

Use the starters and word bank to help you

A rectangular sign holder has a curved top edge.

2 Describe two different ways the curved top could be created in 3D CAD. (3 marks)

Sentence starters

Extrude method: …

Revolve method: …

Word bank

Revolve · method · sketch · fillet · profile · Extrude

Tip: 3 marks = 3 separate, developed points.

YOU DO **On your own** No notes. Check your answers afterwards.

3 Quick check: tick one box for each.

a) For a revolve, which TWO things carry marks?
☐ Colour and material ☐ Centre axis and angle ☐ Fillet and chamfer ☐ Work plane and zoom

b) Joining a large circle to a smaller circle on an offset plane is a...
☐ Revolve ☐ Loft ☐ Extrude ☐ Subtract

c) Which hollows out a solid to a wall thickness?
☐ Fillet ☐ Shell ☐ Loft ☐ Mirror

d) A wine glass stem with a curved profile is best made with...
☐ Extrude ☐ Revolve ☐ Shell ☐ Chamfer

4 Key terms: write the letter of the matching description next to each term.

1. **Extrude**	____	**A** Join two or more profiles on offset work planes.
2. **Revolve**	____	**B** Pull a 2D profile out to a set distance to make a solid.
3. **Loft**	____	**C** Hollow a solid to a wall thickness, removing chosen faces.
4. **Shell**	____	**D** Rotate a profile around an axis; state axis and angle.
5. **Extrude subtract**	____	**E** Remove material using a profile, e.g. for a hole or slot.

A tapered knob is Ø100 at the base, Ø60 at the top and 80 tall. A Ø20 boss 12 high sits on top, with a 4 wide slot across it.

5 Describe how the knob could be modelled using 3D CAD. You may use any suitable method. (4 marks)

A hollow plastic storage box is 120 × 80 × 60 with 3 mm walls and an open top. Its four vertical edges have R10 fillets.

6 Describe how the box would be modelled, in a sensible order. (4 marks)

Check and reflect

Mark your answers with the answer sheet. Go back to the checklist on the site and re-rate each “I can” statement above. Which one still needs work?